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MemoryVR: Collecting and Sharing Memories in Personal Virtual Museums

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ABSTRACT

The development of virtual reality (VR) technology has allowed virtual museums to enrich experiences of personal memories. In this work, we first conducted a survey study to understand the way people use to record, store, and share their digital memories. Informed by the results, we present MemoryVR, a personal virtual museum system designed to preserve and share digital memories. It allows individuals to organize and share their own memories in a virtual museum environment and to visit the memory spaces of others in an immersive way. We invited participants to experience MemoryVR and analyzed their behavior and expectations. The evaluation results showed that users perceived excellent pragmatic and hedonic qualities of MemoryVR. Participants found their experiences of memories in personal virtual museums to be fulfilling. Additionally, we gathered suggestions from participants about MemoryVR, providing design recommendations for customized personal virtual museums.

KEYWORDS

Virtual reality; virtual museum; memory; revisit intention

1. Introduction

The development of digital technology has allowed us to witness dramatic changes in the way memories are recorded and shared. For example, the method of memory recording has realized the transformation from writing, painting, and dictation to photos, videos, and audio. The place where memory recording appears has gradually shifted from diaries, letters, and cave walls to electronic devices, websites, social media platforms, etc. Early in 2006, researchers examined the possibility of developing computer systems and integrating media types that digitally index, store, and retrieve personal memories with the development of Information and Communication Technology (ICT) (O'Hara et al., 2006). Then, customized virtual museums (VMs) have once become popular in the Web 2.0 era (Hayashi et al., 2016; Wang et al., 2009), due to the role that museums play in collecting social and cultural memories (Noy, 2018; Stainforth, 2017). These customized VMs have generally been used by museum enthusiasts to store their favorite artifacts and support them in constructing their unique museums Liang et al. (2024). Over the past decade, advances in 3D graphics and dedicated virtual reality (VR) hardware have spurred the development of immersive and interactive applications (Wang et al., 2024), which have undoubtedly provided novel feasibility for customizing personal VMs. Hence, users' behavior and expectations regarding personal VMs deserve to be explored, since the related research on the design of personal VMs is still limited.

Additionally, this research included a construct *Revisit Intention* (RI) in this study and aimed to explore the factors that may influence it. The phenomenon of user revisits has received continued attention from both academia and industry for its impact on the economic benefits of a destination (Oppermann, 1997), its ability to deepen relationships with museums (Loureiro & Blanco, 2023), its impact on brand loyalty (Jani & Han, 2014), etc. Hence, the ability to attract revisiting users is considered a key factor in measuring core competitiveness (Zhang et al., 2021). Previous research has typically

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explored the impact of RI on a physical location in the real world. It remains unclear what leads to revisiting by users based on VM context, which would allow for the long-term viability of a system. Similarly, how memories collected in personal VMs will translate into revisit intention deserves to continue to be explored. Thus, we investigated the role of autobiographical memory and its relationships with positive emotion and revisit intention. We expected the results to help personal VM designers better identify the design factors that are more appealing to users.

In this paper, we use *personal memories* to refer to subjective, personally meaningful experiences that contribute to an individual's life narrative. Among various mediums of personal memories, such as printed photo albums and handwritten diaries, *digital memories* are those captured and stored through digital technologies (e.g., photographs, videos, social media posts), often used for reflection and sharing. In addition, we draw on psychological literature to define *autobiographical memory* as self-relevant memories grounded in specific times and places, typically rich in emotional and contextual details (Kirkegaard Thomsen et al., 2025), providing a holistic view of an individual's experiences. Unlike digital memories, which can be fleeting and fragmented, autobiographical memories are often holistic and deeply integrated into an individual's understanding of themselves. These distinctions guide our conceptual framing and design considerations throughout the study. We explored the design of a system to expand the practice and research of customized VMs that host personal memories in digital forms. We first explained digital memories from a historical perspective and examined how and why people may treat their digital memories. Then, we examined the definition of VM and the trend in its customization and immersive experience with the development of VR. Based on the theoretical background, we conducted Study 1 that investigated the **RQ1**: *How do users (expect to) use technological means to record, store, share, and revisit memories?* An online survey study ($n = 204$) was carried out to specify user behaviors with digital memories and their expectations for VMs.

After generating some design considerations from the results of Study 1, we designed a personal VM system, MemoryVR, to collect and share personal memories (see Figure 1). Two studies were conducted to understand user experience and behavior when collecting (Study 2, **RQ2**) and sharing (Study 3, **RQ3**) memories in MemoryVR. The results showed an overall good user experience when participants created their own VMs, and an excellent experience when visiting others' VMs. Furthermore, based on the work of Zhang et al. (2021), we found that the photograph and video quality positively influence users' autobiographical memory, which further influences their positive emotion and revisit intention of MemoryVR.

In summary, Study 1 and Study 2 adopt a descriptive approach to explore participants' expectations and experiences with MemoryVR. These insights inform the analytical model tested in Study 3, which examines the relationships between memory quality, emotional response, and revisit intention using structural equation modeling. This staged approach reflects a progression from exploratory insights to analytical validation. Our research contributes to a better understanding of the practical value of the latest research related to VMs and digital memory and promotes a wider adoption of interactive technologies in relevant industries. Specifically, our research has made the following contributions:

- We took a first step to explore the possibility of using customized 3D VMs to collect and share personal memories in digital forms, and demonstrated an artifact design, MemoryVR. Our study showed empirical evidence of its usability and positive effect on the user experience of memories.
- Second, we analyzed and discussed the users' behaviors and expectations during the creation and visiting process. Based on the findings, we give design suggestions for personal VMs in five aspects:

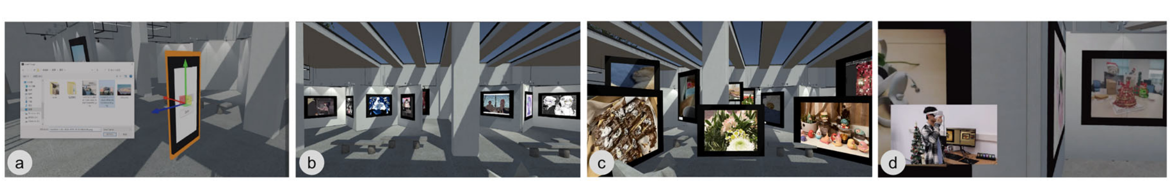


Figure 1. Illustrations of MemoryVR: (a) adding and editing media components of memory moments to the virtual museum; (b-c) screenshots of environments created by participants; (d) a participant visiting a personal virtual museum populated with memory moments using VR.

device selection, access settings, uploaded file types, template provision, and interaction settings. These design suggestions will inspire future design in personal VMs.

- Third, we verified the factors that influence users' revisit intention of MemoryVR. The findings will contribute to understanding how users decide to revisit a personal VM, explaining the indirect relationship between photograph and video quality and revisit intention, and the positive impact of autobiographical memory on positive emotion and revisit intention. Personal VM designers can utilize these findings in the design process to enable users to build autobiographical memories and acquire positive emotions for optimal use of the system.

2. Related work

2.1. Digital memories: Re-assemble, revisit, and share

According to Garde-Hansen et al. (2009), when viewed at a personal level, memory encompasses a broad range of elements from an individual's past experiences, such as particular moments, places, and people, which are carried about in one's mind and contribute to one's self-identity. Over the centuries, people have relied on physical objects as external memory and reference aids, such as diaries, letters, books, and cave walls, which served as repositories of factual information and emotional connections, allowing for the sharing of these with others (Eliseev & Marsh, 2021; O'Hara et al., 2006). However, the advent of digital media technology has ushered in a gradual shift in how people store and record their memories. This shift has evolved from the reliance on physical artifacts to embracing mixed digital and physical environments, and then increasingly turned to digital memories (O'Hara et al., 2006). In our modern context, digital memory encompasses photographs taken with digital cameras or phones, online moments and diaries, computer games, social networking, and many more (Garde-Hansen et al., 2009). It is estimated that Facebook users post more than 200 million new photos every day (Li, 2020). Conceivably, the scale and form of digital memories will expand rapidly with the development of ICT.

Prior work on social media behavior offers useful insights into how users record, share, and reframe personal content across various platforms. For example, Ham et al. (2019) investigated the motivations behind content sharing across multiple social media platforms. It identifies four key dimensions of sharing motivations (i.e., social presence, social conversation, easy connection, and self-management) through focus group interviews and surveys. The study integrated the Uses and Gratifications (U & G) framework, Social Exchange Theory (SET), and the Theory of Reasoned Action (TRA) to understand how these motivations influence sharing behavior. U & G theory explains how individuals actively select media to fulfill specific psychological or social needs, such as emotional expression, information seeking, or self-presentation (Ham et al., 2019). This is particularly relevant in the context of digital memory sharing. Similarly, SET suggests that users weigh perceived benefits (e.g., affirmation, connection) against potential costs (e.g., privacy concerns, effort) when engaging in sharing behaviors (Cook & Yamagishi, 1992). Finally, TRA posits that individuals' behavior is shaped by their attitudes and perceived social norms (Douglass, 1977). Together, these frameworks offer useful lenses for interpreting how and why participants may choose to share personal memories or revisit digital content. However, our research delves into the distinctive ways individuals conceptualize and engage with immersive memory-sharing experiences, particularly in the context of constructing a personal virtual museum. This setting allows users to navigate and narrate their autobiographical memories in a 3D spatial enclosure, fostering a unique interplay between memory, identity, and storytelling that transcends traditional digital sharing methods.

Given that the formats and amounts of digital memories have increased dramatically, more methods need to be applied to re-assemble and revisit our digital memories through the masses and their associated technologies (Garde-Hansen et al., 2009). Frederic Bartlett, who greatly influenced the psychology of memory, has stated that the key process of memory involves the past into the present to produce a "re-experience" of consciousness (Bartlett, 1933). People may receive benefits from looking back on their past activities, while how much they will enjoy reminiscing about their past is underestimated (Zhang et al., 2014). Professionals have tried to explore opportunities in digital memories' capacity in re-assembling and emotional re-experiencing/reminiscence (Schoenebeck et al., 2016). One typical application that tries to help people to re-assemble and reminisce about memories is Pensieve (Peesapati et al., 2010), aiming to support everyday reminiscence by allowing users to record, upload, and organize their daily memories in different formats, such as photos and texts. According to Pensieve's user research, their participants reported that they valued this personal space to record and re-assemble things in the way they want; when they looked back on those digital memories, generally, they experienced a better mood. At the same time, other positive effects of reminiscing were listed, such as evoking delightful memories, working through one's past, and using what they learned from the past to solve present issues. Apart from these findings about re-experiencing, Pensieve and other likely applications also attach great importance to another function - sharing.

Unlike memories kept in people's minds, memories in digital formats have a stronger potential to be shared, which is in line with a fundamental human need - the need for sharing different kinds of memories (Olsson et al., 2008). Many researchers have examined that through sharing memories, people can convey information about who they are to others, entertain others, and develop and maintain social bonds (Rimé, 2007; Stone et al., 2022). One typical example is the rapidly growing user penetration of Facebook-like applications (Chaffey, 2016), although "face-to-face" conversation is still undeniably important in daily life. This reflects the fact that sharing digital memories can somewhat meet individuals' needs.

2.2. Virtual museums: Towards customizable and immersive experience

"Virtual Museums" (VMs) are an evolving construct. Some scholars referred to museums as "memory institutions" because of their role in social and cultural memory (Stainforth, 2017), while others defined a broader scope, considering it as a collection of all information, reflecting changes influenced by developments in Information and Communication Technology (ICT) (Schweibenz, 2019).

Despite museums being often associated with collective memory (Noy, 2018), O'Hara et al. (2006) recognized the potential for museums to enhance the development of personal memory systems in the context of the rapid development of digital media technologies. Traditionally, museum artifacts are organized according to inherent classification systems, which may not fully address the divergent needs of visitors. With the development of ICT, the constraints are gradually eliminated in the development of VMs (Iqbal et al., 2021). Hayashi et al. (2016) suggested that the era of "Web 2.0" introduced the idea of customizable web content, sparking increased interest in research concerning customizable VM services tailored to individual users. More recently, the growing spread and acceptance of VR technology have shed light on VM research (Lee et al., 2022) and opened up new possibilities for the research agenda. VR has also been seen as a potential way to customize individual experiences.

Shehade and Stylianou-Lambert (2020) claimed that people can easily build an inner emotional connection through a VR environment. Early VMs were characterized by multimedia and hypermedia applications on CD-ROM and stand-alone computers (Schweibenz, 2019). However, the conventional 2D interfaces restrict the user experience to basic page viewing and sequential browsing, limiting opportunities for immersive experience (Iqbal et al., 2021). Until more recently, immersive VMs have been made possible and more accessible along with the recent VR hardware and software development (Lee et al., 2022). Unlike interactions through conventional computing devices such as computers or mobile phones with rectangular displays, VR eliminates the edges of the screen but provides a fully immersive display (Bucher, 2017). In this case, VR transcends physical limitations and enables users to explore new worlds beyond their situated location and time (Guerra et al., 2015). The strong "sense of being there" afforded by VR also fosters deeper and broader emotional responses (Peng et al., 2020), compared to traditional media and devices used for VMs.

Another line of relevant research involves the use of immersive virtual environments for mnemonic purposes, such as the *method of loci* or *memory palaces* (Krokos et al., 2019; Smith, 2019; Smith & Mulligan, 2021). The *method of loci* is a mnemonic technique that involves visualizing a familiar spatial layout to enhance memory recall by associating specific pieces of information with distinct locations within that space. In contrast, a *memory palace* refers to a more elaborate application of this method, where users create a detailed imaginary structure (like a palace) filled with vivid imagery and associated information, allowing for more complex and personalized memory associations. For example, Krokos et al. (2019) explored how users can upload and view media within immersive 3D environments, using spatial memory for enhanced recall. They found that immersive VR memory palaces significantly improved recall accuracy compared to traditional desktop-based methods. Similarly, Mastrogiorgio et al. (2021) investigated whether virtual memoryscapes can counteract digital amnesia in organizations. Researchers compared the efficacy of virtual memoryscapes with the traditional method of loci, discovering that the method of loci was more effective in the short term, but virtual memoryscapes performed equally well in the long term without significant memory decay. These works demonstrate the potential of VR to support structured memory encoding through spatial immersion. While conceptually related in their use of immersive space, these mnemonic applications differ from our goal. Rather than merely supporting the *recall* of abstract information, we focus on autobiographical memory, which encompasses emotional engagement and personal meaning-making. This distinction highlights that memory, in our context, is not just about memorizing facts or data; it is about the rich tapestry of personal experiences, feelings, and reflections that shape an individual's identity and understanding of their life journey.

Situating our work in the interdisciplinary area of digital memories and virtual museums, the review of related work demonstrated the strong potential of using personal virtual museums for immersive experiences of memories, contributing to the construction, customization, revisit, and sharing of users' digital memories, and facilitating their emotional connections.

2.3. Revisit intention: A theoretical framework

Numerous studies have given different definitions of revisit intention, such as the desire to visit a previous destination within a specific timeframe (Cole & Scott, 2004), the extension of satisfaction (Um et al., 2006), customers' loyalty with products (Jani & Han, 2014), etc. Zhang et al. (2021) interpreted revisit intention as the tendency to return to the same destination. They proposed a conceptual framework on tourists' memory experience, exploring factors that would influence tourists' revisit intention based on the Stimuli-Organism-Response (SOR) framework (Mehrabian & Russell, 1974). Their results demonstrated a significant correlation between travel photographs and autobiographical memory, and a significant positive effect of autobiographical memory and positive emotion on revisit intention. Inspired by their work, we attempt to understand users' revisit intention of personal VMs. Specifically, we build on the groundwork from our descriptive research and adopt an analytical approach that measures users' revisit intention to assess user satisfaction with MemoryVR.

Unlike Zhang et al. (2021), our model omitted two constructs related to place attachment (place identity and place dependence). This decision was based on the fact that MemoryVR represents a digital system rather than a physical destination (Proshansky et al., 1983). In a physical context, place identity refers to the emotional and symbolic significance that individuals attach to a specific location, while place dependence involves the reliance on a location for fulfilling certain needs or activities. However, in the case of MemoryVR, the virtual environment was intentionally designed as an abstract space, devoid of the rich contextual information and physical characteristics typically associated with real-world places. It prioritizes the exploration of personal memories and emotional engagement, free from the constraints of traditional place-related constructs. This allows users to interact with their memories in a way that transcends physical attachment, focusing instead on the emotional and narrative aspects of their experiences. We also excluded the construct concerning the quantity of photographs and videos, as this variable was controlled and kept consistent within the digital system. Thus, we defined and elaborated four constructs: photograph and video quality, autobiographical memory, positive emotion, and revisit intention.

2.3.1. Autobiographical memory

As a core part of the self-memory system, autobiographical memory is generated by an individual's personal experience (Conway & Pleydell-Pearce, 2000; Kim, 2014). It refers to a subset of personal memory that involves consciously recalled events with personal significance. Kirkegaard Thomsen et al. (2025) described autobiographical memory as part of a broader memory ecology that includes personal, vicarious, and collective memories. In this view, autobiographical memory is tightly linked to identity formation, narrative structure, and emotional engagement. These distinctions are important for understanding how MemoryVR may influence users' sense of self and the emotional salience of revisiting digital content.

2.3.2. Photograph and video quality

It refers to the uniqueness, interest, and affluence of the media. Users use photographs and videos to establish relationships with other people and to narrate. Wilhelm et al. (2004) noted that photography had three main social functions: memory awakening, relationship maintenance, and self-expression, with memory awakening being seen as the most significant. The higher the quality of the photographs and videos, the stronger the visual impact on the audience, and the greater the autobiographical memory evocation (Kim, 2010; Zhang et al., 2021). Thus, we propose

H1 Photograph and video quality have a positive impact on users' autobiographical memory.

2.3.3. Positive emotion

Positive emotion is vital in people's daily life, assisting people to retain a healthy mental state (Diener & Seligman, 2002). It indicates a pleasurable feeling that arises from meeting an individual's needs. Previous research showed that autobiographical memory evocation is often accompanied by positive emotion (Baumgartner et al., 1992). Hence, we propose

H2 Autobiographical memory contributes to positive emotion.

2.3.4. Revisit intention

Revisit intention indicates users' willingness to visit the same content again. Autobiographical memory and positive emotion were found to have a positive impact on users' tendency to visit the same system again (Martin, 2010). Thus, we propose

H3 Autobiographical memory contributes to revisit intention.

H4 Positive emotion contributes to revisit intention.

The proposed research model is illustrated in Figure 2.

3. Study 1: Investigating the recording, storing, and sharing of digital memories

Study 1 was conducted to answer **RQ1**: *How do users (expect to) use technological means to record, store, share, and revisit memories?*

3.1. Survey design

A survey was designed and distributed online, consisting of 34 questions. Apart from the four questions (Q2-Q5) collecting respondents' basic demographic information (gender, age, job, and educational background), the main body of the survey contains three sections.

In Section I, we included 15 questions (Q6-Q20) to investigate how users tended to record and store their digital memories. Multiple-choice questions and frequency questions were used, such as Q6: *How*

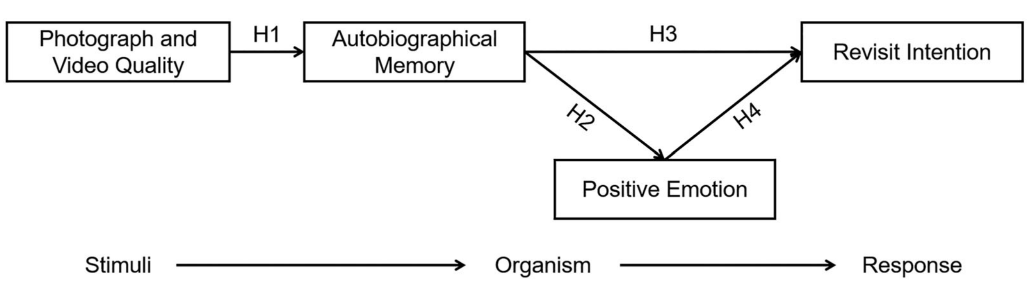


Figure 2. The research model about user revisit intention.

do you like to record your life?, Q9: *How often would you record your memory moments?*, and Q13: *How satisfied are you with your current storage of personal memory?* We also included open-ended questions to ask follow-up explanations, such as Q14: *Can you briefly explain what you are not satisfied with?*

Section II had 6 questions (Q21-Q26), aiming to verify if they were interested in sharing their own digital memories and exploring others' memories. For example, *How often do you share your selected life moments online* [Q21: *publicly*, Q22: *with close friends, family, and colleagues*]? and *How often do you view* [Q23: *others'*, Q24: *your close friends, family, and colleagues'*] *selected life moments online?* We provided examples of selected life moments, e.g., texts, photos, videos, and examples of popular online platforms, e.g., WeChat, TikTok, Little Red Book, LinkedIn, Facebook, etc.

Section III served to understand if it is appropriate and if users are willing to revisit digital memories using VR technology, using 8 questions (Q27-Q34). Specifically, Q27-30 asked about users' VR experience, knowledge, and the VR devices they have. They were asked to evaluate their interest and willingness to learn new technologies and build their own personal museums, as well as the most likely motivation for building a personal museum in Q31-34. More details about the survey can be found in Appendix A.

3.2. Participants

In total, we received 204 complete responses (106 females, 96 males, and 2 non-binary), with a sample age between 16 and 53 ($M = 27.81, SD = 9.36$). 76% of the respondents ($n = 155$) have a bachelor's degree or above, and around half (46.60%) of respondents ($n = 95$) are students. Based on the IP address data at the time they filled in the survey, 4.41% percent of respondents ($n = 9$) lived in developed countries such as the United Kingdom, the United States, and some countries in Europe. We examined their responses and found that these were all filled out in Mandarin Chinese. The remaining 95.59% of the respondents ($n = 195$) lived in China. More than 69.12% of them ($n = 141$) are from the Yangtze River Delta Region of China, which is regarded as a region with relatively developed economic development in China. Even though most of the respondents ($75.48\%, n = 154$) are from areas with relatively high-level development, the adoption of VR devices is still far from ubiquitous. The results showed that 64.71% ($n = 132$) of the respondents have prior experience in using VR (Q27), but only 26.47% ($n = 54$) of the respondents have their own VR equipment (Q29). Most participants were slightly familiar with VR (Q28, $M = 2.53, SD = 1.14$).

3.3. Analysis and results

3.3.1. Recording and storing of digital memories

Compared with using non-digital methods, such as on a notebook ($11.76\%, n = 24$), more participants prefer to use digital devices such as smartphones ($88.24\%, n = 180$), cameras ($34.80\%, n = 71$), and PCs and laptops ($34.31\%, n = 70$) to record life moments (Q6). Consequently, viewing them on digital devices ($94.12\%, n = 192$) is more popular than non-digital methods (Q10). Besides, more participants are used to keeping their digital memories in the forms of images ($81.86\%, n = 167$) and videos ($77.94\%, n = 159$) compared to text ($46.57\%, n = 95$) and voice ($28.92\%, n = 59$) (Q7). Q20 showed that they also prefer to include their digital memories in the form of videos ($86.27\%, n = 176$) and images ($80.88\%, n = 165$) in a 3D virtual space (see Figure 3).

As for what they recorded in their lives (Q8), travel has the highest frequency ($76.96\%, n = 157$), followed by food ($66.18\%, n = 135$), close relationship ($59.31\%, n = 121$) and pets ($56.37\%, n = 115$). Participants also record their achievements and outdoor activities ($55.39\%, n = 113$), and lastly their work ($49.51\%, n = 101$) and performances they watched ($25.98\%, n = 53$). The majority of participants ($72.55\%, n = 148$) recorded their memory moments more than a couple of times a month (Q9).

3.3.2. Sharing of digital memories

The frequency of participants sharing their own life moments (Q21 and Q22, $M = 2.49$ and 3.00, $SD =$ 0.94 and 1.11) is less than the frequency of viewing others' life moments (Q23, $M = 3.47, SD = 1.23$, Q24, $M = 3.52, SD = 1.24$). Specifically, they tend to share life moments with close friends, family, and colleagues ($M = 3.00, SD = 1.11$) rather than publicly with strangers ($M = 2.49, SD = 0.94$). Similarly,

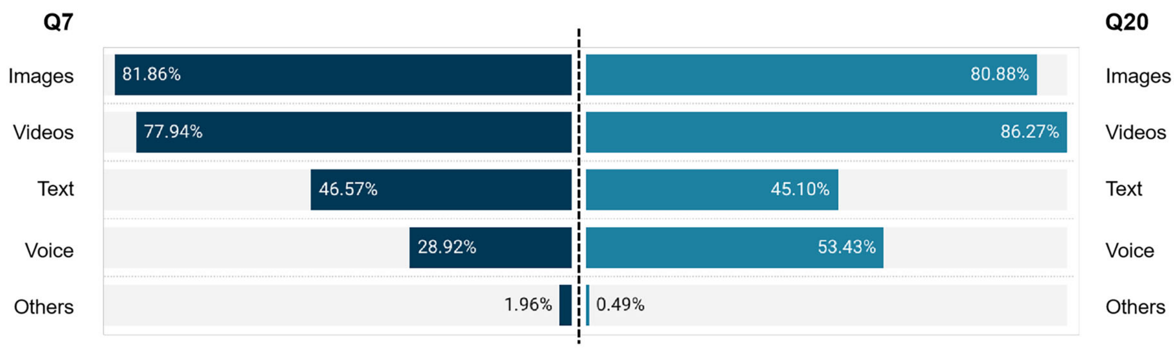


Figure 3. Results of Q7: *What kinds of digital memory formats would you like to keep?* and Q20: *If you were asked to create a personal museum (a 3D virtual space) that hosts your memory, what kind of media would you like to include?.*

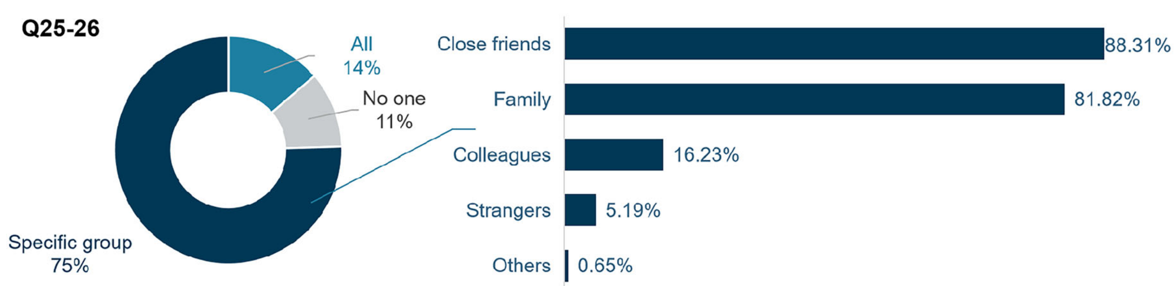


Figure 4. Results of Q25-Q26: *If you were asked to create a personal museum (a 3D virtual space) that hosts your memory, who would you like to share this personal museum with?.*

when they were asked about sharing their life moments hosted in a personal virtual museum (Q25-Q26), 75.49% of participants ($n = 154$) claimed that they are more willing to share with specific groups such as close friends ($88.31\%, n = 136$), and family ($81.81\%, n = 126$), whereas only 13.72% ($n = 28$) chose everyone and 10.78% ($n = 22$) chose no one (see Figure 4).

Overall, participants are moderately satisfied with the current storing (Q13), viewing (Q15), and sharing (Q17) of personal memory, with an average scale (rated from 1 = Not at all satisfied to 5 = Extremely satisfied) of 3.36 ($SD = 0.83$), 3.44 ($SD = 0.81$), and 3.52 ($SD = 0.72$), respectively. As participants reflected in Q14, Q16, and Q18, reasons for unsatisfactory experience include the lack of overview (e.g., *can't comprehensively show what I have done*), being difficult to find, search, and classify (e.g., *messy*, *unorganized*), and having *limited forms of sharing*. However, participants generally have a positive attitude towards having a 3D virtual space where photos and videos can be stored (Q19, $M = 3.37, SD = 1.03$).

3.3.3. Revisit digital memories

Most participants showed positive feedback regarding their interest in experiencing new technologies (Q31, $M = 3.22, SD = 1.16$) and willingness to invest in learning new things (Q32, $M = 3.95, SD = 0.84$). Similarly, most of them (86.76%, $n = 177$) showed a positive or neutral willingness to create a 3D virtual space that hosts a collection of digital memories, with an average willingness of 3.63 ($SD = 1.00$) (Q33). In the sorting question Q34, 85.78% ($n = 175$) of participants regarded being able to revisit their moments of life as the first motivation to build a personal museum (a 3D virtual space) that hosts memory, followed by storage ($69.12\%, n = 141$) purposes and social ($55.39\%, n = 113$) needs.

3.4. Summary of findings

The survey results give answers to **RQ1**, indicating that (1) most people are accustomed to using digital devices to record, store, and review their life moments in the form of images or videos; (2) most people have frequent needs to view others' life moments and to share their own life moments with others; (3) while they do share selected life moments publicly with strangers, the majority tend to expect having a personal virtual museum shared with close friends and family only; (4) users are interested in building an immersive personal museum that hosts their digital memories, primarily motivated by its affordance in re-experiencing their past; and (5) nowadays, people having their own VR devices is still a minority.

4. Design and implementation of MemoryVR

Based on the review of related work and the survey results from Study 1, we identified several design priorities for MemoryVR, a personal virtual museum system (see Figure 1).

4.1. Design considerations

4.1.1. Functions

Study 1 identified two major functions to support. (1) *Create and Customize*. Users should be allowed to create and edit a new VM, including uploading images and videos they want. (2) *View and Share*. The system should allow users to share their furnished VM and visit VMs shared by others.

4.1.2. Constraints

Unlike VMs that only allow users to navigate and explore a pre-defined space, personal VMs engage users to create their own environment and memory experience. Although VR was found to allow the revisit of memory in an immersive way, the creation of 3D VM layouts in VR can be challenging for novice users, and based on our research, many people do not own a VR device.

4.1.3. Devices

Based on the desired functions and existing constraints, we conclude that it is necessary to include both PC and VR for the personal VM system. The system supports users to arrange their digital memories in VMs using a PC and visit the VMs created by themselves and others. For users with VR devices, the system supports them in connecting the VR devices to visit their personal VMs and have an immersive memory experience.

4.2. Hardware, software, and development

The MemoryVR system was developed using Unity (version 2020.3.21f1). The system supports the use of personal computers and the Meta Quest 2 VR headset with two controllers. To record the users' account information and the personal VMs they created, we created databases using *phpMyAdmin* and connected them to Unity.

4.3. Interaction design

4.3.1. Create by PC

For personal VM creation, users were supplied with a standard PC setup, including a monitor, mouse, and keyboard. Four actions are supported: (1) **Move**. Users can press `WASD` keys to move forward, left, backward, and right; (2) **Change viewing angles.** Users can press and drag the mouse to adjust the viewing angle; (3) **Upload**. By pressing the `Esc` key to exit the view and trigger the menu, users can click on the `Add Model` button to add a *Photo* frame or a *Video* frame. It allows users to upload `.jpg` and `.png` images and `.mp4` videos; (4) **Edit media frame.** By clicking on the media frame, users can press `Q`, `E`, or `R` keys to enable the movement, rotation, or scaling of the media frame (see Figure 5).

4.3.2. Visit by PC

With the same operations in the *Create* process, users can (1) **move** in the VM environment and (2) **change viewing angles**. In addition, users can (3) **play/pause video** by clicking on the play icon on the video frame; and (4) **take snapshot** using `Alt` + `A` keys.

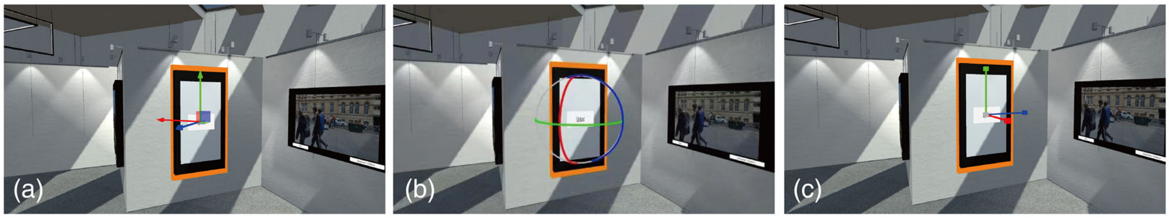


Figure 5. Editing of the media frame: (a) move, (b) rotate, and (c) scale.

4.3.3. Visit by VR

Users could visit personal VMs using a VR HMD, such as Meta Quest 2. The four actions in the PC visit process are also supported in VR. Both continuous and discrete locomotion techniques are allowed (Zhao et al., 2023) for users to (1) **move** in the scene. By turning their body or the right controller's joystick, they can (2) **change viewing angles**. Users can use controllers to touch (collide) the play icon on the video frame to (3) **play/pause video**. In addition, they can grab the camera in the scene and press the trigger to (4) **take a photo**.

5. Study 2: Collecting memories in MemoryVR

This study was designed to investigate **RQ2:** *How do users experience and behave when creating their own memories in MemoryVR?* We conducted an exploratory study and invited participants to create their own VMs, as well as evaluate the system usability and user experience with MemoryVR. Participants were recruited using a snowball sampling technique (Goodman, 1961), as the findings from Study 1 suggested that users were inclined to share their memories with close friends. This study focused on the creation process using a standard PC setting, including a monitor, mouse, and keyboard.

5.1. Measurement and scoring

5.1.1. Usability and user experience

System usability was measured by the System Usability Scale (SUS) (Brooke, 1996). It consists of 10 questions scored on a five-point Likert scale ranging from 1 (strongly disagree) to 5 (strongly agree). After a negation of the even-numbered questions, the usability score can be calculated by $\sum_{i=1}^{10}(sus_i - 1) * 2.5$, with 68 being the threshold value of average usability.

User experience was measured by the short version of the User Experience Questionnaire (UEQ-S) (Schrepp et al., 2017). It comprised two sub-scales, namely pragmatic and hedonic quality, along with a total score that reflects users' holistic impression of the system. UEQ-S consisted of 8 items with scores ranging from −3 (horribly bad) to 3 (excellent). A score between −0.8 and 0.8 represents a neutral evaluation, and a score greater than 0.8 indicates a positive evaluation.

5.1.2. Comments on MemoryVR

We used two methods to collect participant comments. First, we encouraged participants to conduct the experiment by simply expressing their thoughts by adopting a Think Aloud approach (Duncker & Lees, 1945), a usability approach that could yield useful qualitative feedback in HCI practices (Nielsen et al., 2002). Second, we conducted a semi-structured interview with the participants after the experiment to explore their experience further. The interview consisted of two questions: **Q1:** *Does MemoryVR give you a different experience compared to other ways of sharing personal memories?* **Q2:** *Do you have any other comments or requirements about the MemoryVR system?*

We audio-recorded the entire experiment process. Automatic transcription and translation services were used to transcribe the recordings into texts and translate them into English. The automated transcription results have undergone manual correction and verification, and the same process has been applied to the automated translations. The qualitative data were coded and categorized using NVivo (version 12 plus), and analyzed using the theme-based content analysis (Neale & Nichols, 2001). This approach provided indications for qualitative data results by categorizing the data into meaningful categories.

5.2. Procedure and tasks

Participants voluntarily signed up for the study, knowing that they needed to prepare images and videos to create a personal VM. Upon their arrival at the laboratory, we clarified the purpose of the study and introduced the experimental procedure. Participants were told that the data they used to create the virtual museum would be stored in the database. After participants signed the informed consent form and filled in the demographic questionnaire, they were asked to select and copy the images and videos on the experiment computer. We suggested participants include at least five images and one video, but no upper limit. They were provided with a tutorial manual with screenshots and annotations that introduce the operations described in Section 4.3.1. They could start the creation once they were ready.

During the PC creation process, participants started from a standard gallery setup (see Figure 1) with no media frames placed in the scene. Participants were asked to complete two tasks: (1) create a personal VM, upload the prepared images and videos, and edit the layout; (2) name their personal VM and indicate if it can be viewed by others. After completing the tasks, participants filled in a questionnaire to evaluate the usability and user experience. At the end of the experiment, participants attended a semi-structured interview lasting approximately 5 min to collect their comments about MemoryVR. We also asked them if they could invite friends to participate in the study. The experiment lasted 45 to 50 min in total. The study is approved by our University Ethics Committee. Figure 6a illustrates the procedure of the study.

5.3. Participants

A total of twenty participants (11 females and 9 males) aged between 18 and 26 ($M = 21.90, SD = 2.51$) took part in the PC *Create* study. Our participants were undergraduate ($n = 13$) and postgraduate students ($n = 7$) from a local university. Participants were very familiar with PC ($M = 4.05, SD = 1.05$) and familiar with 3D graphics software ($M = 3.05, SD = 1.15$). They reported that they enjoy collecting and sharing their personal memories ($M = 3.45, SD = 0.69$).

5.4. Analysis and results

5.4.1. Usability and user experience

The mean SUS score of the PC creation in MemoryVR was 71.50 ($SD = 11.85$), which exceeded the threshold value of 68.00, indicating satisfactory usability of the MemoryVR system.

The UEQ-S results also showed positive ratings (see Figure 7a). The pragmatic quality, hedonic quality, and overall experience had mean scores of 1.33 ($SD = 0.98$), 1.75 ($SD = 0.93$), and 1.54 ($SD = 0.75$), respectively, exceeding the threshold value of 0.8 (Laugwitz et al., 2008). The results

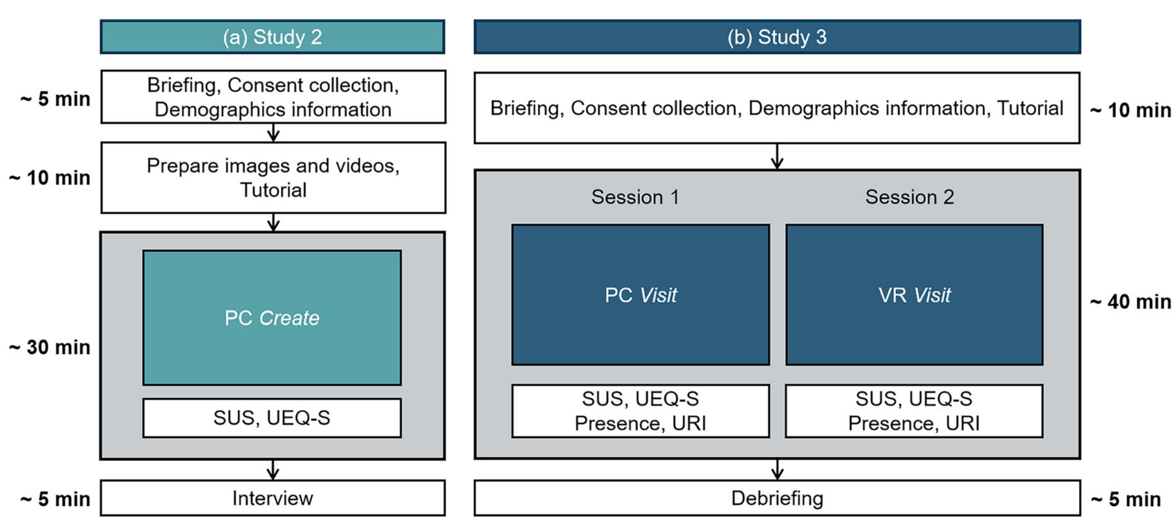


Figure 6. Example experimental procedure of (a) Study 2 and (b) Study 3.

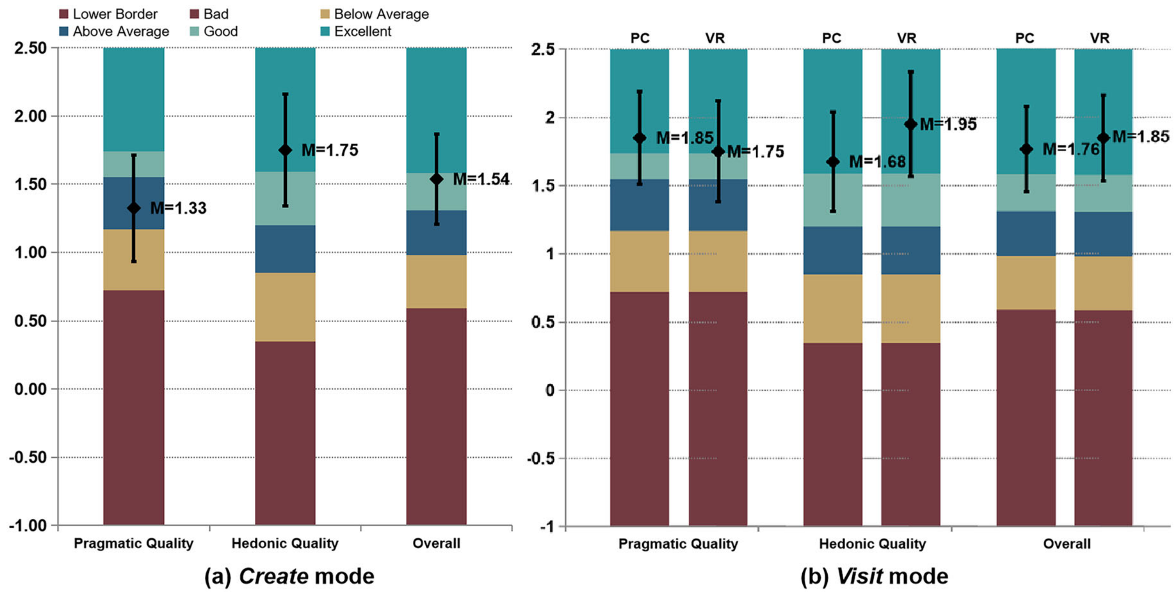


Figure 7. The results of UEQ-S scores from (a) Study 2, PC create and (b) Study 3, PC and VR visit.

indicate that users had a positive experience with MemoryVR. Compared to the UEQ benchmark data-set (Laugwitz et al., 2008), the pragmatic quality of user experience was *Above average* (top 50% to 25%), the hedonic quality was *Excellent* (top 10%), and the overall experience was *Good* (top 25% to 10%).

5.4.2. Time spent, content in VMs, and naming of VMs

5.4.2.1. Time spent. We recorded the time they spent creating their personal VMs and the number of media frames uploaded in the 20 VMs. The create time they used varied from 783s to 1842s ($M = 1285.40, SD = 283.35$).

5.4.2.2. Number of images and videos. The number of images included in the VMs ranges from 5 to 23 ($M = 10.80, SD = 5.30$); the number of videos ranges from 0 to 5 ($M = 2.55, SD = 1.15$). Even though we asked users to upload at least one video, there were still two users who did not upload any video. Participant (CP9) explained that "I never record videos, so I can't find any video in my album."

5.4.2.3. Contents and themes in VMs. Many participants ($n = 12$) created their VMs based on a specific theme and selected content around that theme, such as pets (CP2, CP8), food (CP6), concerts (CP14), musicals (CP15), sea (CP17), etc. In contrast, others uploaded various types of images and videos related to themselves. We coded and analyzed the 216 images and 51 videos uploaded by participants, which can be categorized into eight types (see Figure 8). Some photos could be categorized into multiple types (such as), so the total counts exceeded the number of images and videos. Different from the results in Study 1 (see Section 3.3.1), images and videos about their close relationship consisted of the largest number of uploads in VMs ($n = 71$), followed by travel ($n = 60$), decoration ($n = 47$) and their pets ($n = 46$).

5.4.2.4. Naming of VMs. Twelve participants named their VMs by describing the theme (e.g., *Did you see the concert today?*), six participants used their nicknames or initials (e.g., *XM's museum*), and two participants named their museums using meaningful numbers that represent codes of a place or a person. Many participants considered intuitive names for potential visitors. In other cases, participants chose to use their real names or nicknames. This indicates the identity associated with their personal VMs. This also appeared to be the default choice when they had few name ideas, as CP3 indicated, "I'm not sure what name to use, so I just typed my own name."

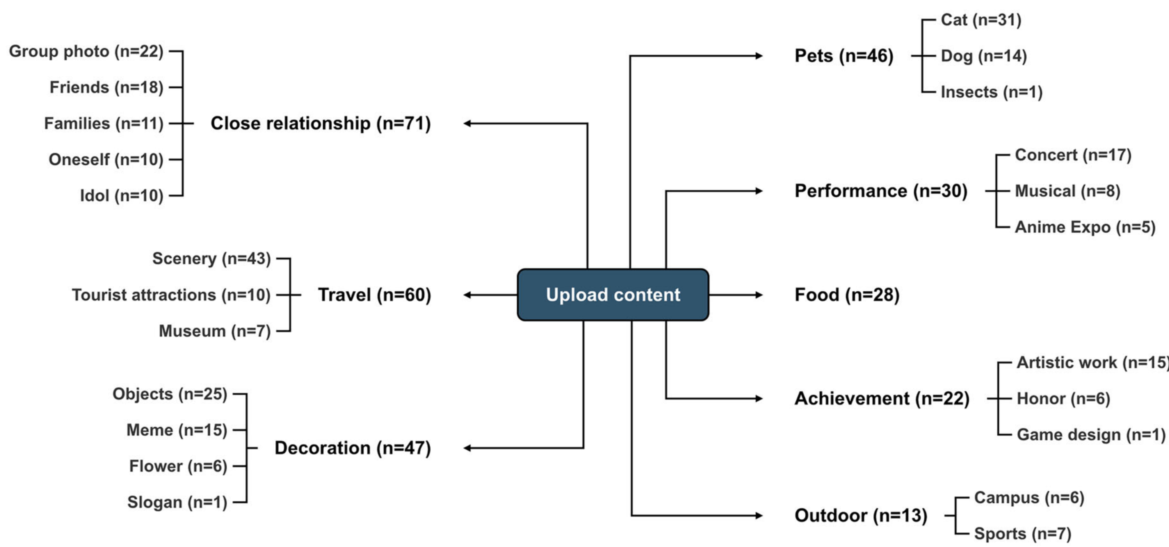


Figure 8. The content uploaded by the users was classified into eight categories, as well as the number of items included in each category.

5.4.2.5. Public or private?.
As for the sharing access of their personal VMs, CP1 and CP3 indicated that they only wanted to share with families and friends, and the others were willing to publicize them.

5.4.3. Interview results

Compared to common ways of storing memories (such as mobile phone albums, social media, etc.), MemoryVR was found to be a novel approach. Customized personal VMs gave users a better experience in storing memories. CP14 stated that "other platforms are (of) flat (displays). This system makes me feel like I'm really creating a personal museum, and it's owned by me. The feeling is kind of amazing." According to CP14, MemoryVR is more multidimensional, making her creative process more realistic. The comment from CP6 complemented this opinion and mentioned the viewing experience. "Traditional social media is on a flat screen, and I can only settle on one interface to swipe through. The direction of swiping is also fixed, just up and down. However, in this kind of museum, I have free actions so that I can explore in different ways." CP6 highlighted that the sense of "freedom" in MemoryVR brought him a great experience.

Regarding the use of MemoryVR for sharing, the majority of participants expressed a willingness to be public. However, CP7 mentioned the impact of access settings on his content selection, "I did pick my content knowing that what others will see my VM." Besides, participants mentioned that they recalled and reorganized their memories when creating their personal VMs, which rarely occurred when sending social media posts. "I especially liked the process of recalling when creating my own VM. It was great to review my experiences and then display some of my achievements," said CP9. CP12 also echoed this nostalgic feeling that MemoryVR delivers.

5.5. Summary and discussion

Here, we discuss some interesting phenomena that were observed during the experiment, and some suggestions for improvements reported by participants.

5.5.1. Participants' behavior when creating in MemoryVR

We observed that personal VMs tend to stimulate the **ritual sense** of storage and sharing. One of the creative participants (CP12) placed a number of honorary certificates in the VM, and he described his competition experience to the researchers during the creation process. He commented that "I uploaded some of my achievements to this museum to share with other people. The system allowed me to relive the memories I had." In addition, CP10 chose to upload and arrange her most proud digital artworks in MemoryVR during the experiment. She commented that "I love it. I've organized an art exhibition for myself for the first time! It's impossible for me to have an exhibition in real life, but I can invite my friends to visit this virtual one." While this behavior was not exclusive to MemoryVR, it aligns with broader trends in the use of virtual environments for art exhibitions and personal expression, where many platforms enable artists to create immersive experiences that reach global audiences, bypassing barriers such as financial constraints and logistical challenges (Sylaiou et al., 2024). Providing a spatial layout for users to host and revisit memories in their personal VMs can serve as motivation for them to organize their memorable moments and gain a comprehensive perspective on their significant life events, thereby fostering a sense of accomplishment. The transformative potential of virtual environments could also enhance accessibility and foster community, allowing users to engage with their creative identities in meaningful ways.

5.5.2. Participants' expectations of MemoryVR in the creation process

According to participants' feedback, we see three potential improvements in the creation process. The first is to include different environment layouts and increase the customizability of VMs. CP13 pointed out that he hopes that the personal VMs can provide customized environments. "I would love to record a match I had in middle school, is it possible to set a football field scene? I can recall the memory when I see the scene." Similarly, some users were slightly disappointed that they could not change the default layout of the gallery and some properties of the media frames, such as modifying the color of image frames, adjusting the position of the walls, adding some decorations, etc. While most participants positioned media frames on the wall (see Figure 1b), one participant (CP5) chose to completely ignore the default layout of the gallery and created an enclosed space using multiple media frames floating in the air (see Figure 1c). This signifies their experimentation with the affordances of the virtual environment and highlights the benefits of flexibility for customizing their personal VM creations. Secondly, supporting multiple upload types. According to the results of Study 1, MemoryVR supports users in uploading photos and videos. However, in practice, many users indicated that sound and 3D models were also important elements they wanted. Even though it may require higher learning costs (e.g., how to make their own 3D models), they still want to enrich personal VMs with these elements. The third one is to optimize the 3D operations. It is conceivable that individuals who are not familiar with 3D software may encounter challenges in the 3D operations during creation. However, during the experiments, several participants with prior 3D modeling experience noted discrepancies in certain operations compared to what they had previously learned, such as the key used for moving objects (`W` or `Q`). The design choice was primarily made to resolve conflicts between commonly used keys for object editing and those for navigation. Participants anticipated 3D operations to align more closely with their prior experiences. Nevertheless, even software like Unity and Autodesk Maya, as mentioned by participants, do not maintain external consistency in their 3D operations. Clearly, a more effective solution is required to prevent mismatches with users' mental models. Perhaps an alternative is to allow users to directly edit the media and their layouts in VR.

6. Study 3: Sharing memories in MemoryVR

In this study, we attempted to develop an understanding of **RQ3a**: *How do users experience and behave when visiting others' memories in MemoryVR?* and to answer **RQ3b**: *What factors influence users' revisit intention of MemoryVR?* We invited participants from Study 2 to rejoin the study and forward the invitation to their friends if they wished to do so. All participants visited VMs through both a standard PC setting and an HMD-based VR setting. The sequence of using the two devices was counterbalanced following a Latin-Square design to mitigate the sequence effect, where half of them used PC first, and the other half used VR first.

6.1. Measurement and scoring

6.1.1. Usability and user experience

Same as the measurements in Study 2, we asked participants to evaluate the system usability and user experience using the System Usability Scale (SUS) (Brooke, 1996) and the short version of the User Experience Questionnaire (UEQ-S) (Schrepp et al., 2017). The scoring is also consistent (see Section 5.1.1).

6.1.2. Presence

As we included VR during the visit process, we measured users' perceived presence using a question "I have a sense of being there" (Slater, 1999). The responses were recorded on a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree).

6.1.3. User revisit intention

User revisit intention was measured by a group of questions modified from the research by Zhang et al. (2021). It includes thirteen questions that measure four constructs (see Table 1). There were three items for *Photograph and Video Quality* (Zhang et al., 2021), four items for *Autobiographical Memory* (Ely & Mercurio, 2011), three items for *Positive Emotion* (Loureiro, 2014), and three items for *Revisit Intention* (Kim et al., 2015). The responses were recorded on a 7-point Likert scale, ranging from 1 (strongly disagree) to 7 (strongly agree). For simplicity, we abbreviated the measurements of user revisit intention as URI.

To examine the robustness of the questionnaire items, the measurement model was assessed by examining the construct reliability, convergent validity, and discriminant validity. Specifically, the Cronbach's alpha (CA) and composite reliability (CR) values are examined for all constructs to ensure the internal reliability of items (Fornell & Larcker 1981). Convergent validity was then assessed by measuring the average variance extracted (AVE) of factors. The structural model was then evaluated using path coefficients, R^2, and significance values (Hair et al., 2011).

Additionally, to evaluate the proposed hypotheses, we applied Structural Equation Modeling (SEM) using SmartPLS 4. SEM is suitable for testing theoretical models involving multiple latent constructs and allows for the simultaneous estimation of direct and indirect effects between them (Monteiro et al., 2022). In this study, partial least squares (PLS) regression was used to perform bootstrapping for our research model and to test and validate the proposed model, as well as the relationships among the hypothesized constructs (Hair et al., 2011). PLS regression is one of the most commonly adopted structural equation modeling (SEM) techniques used to validate structured data.

6.1.4. Comments on MemoryVR

Similar to Study 2, we encouraged participants to think aloud when interacting with the MemoryVR system. The data was analyzed using the same qualitative analysis methods (see Section 5.1.2).

6.2. Procedure and tasks

The experimental procedure of Study 3 is similar to Study 2. We provided a briefing of the study and collected participants' consent and demographic information. A tutorial is given for the use of PC visits and VR visits of MemoryVR. A tutorial manual with screenshots and annotations was provided to ensure that consistent instructions were given to all participants.

The experiment included two sessions, requiring each participant to use PC and VR, respectively, to visit the VMs that participants created in Study 2. Participants were asked to complete two *Visit* tasks: (1) browse the list of VMs and visit any three of them or more using PC; (2) browse the list of VMs and visit any three of them or more using VR. Although we had twenty VMs in total stored in the database, two participants (CP1 and CP3) indicated limited access to friends, so these two were only shown to their invited participants, but not the others. We followed the Latin Square design to minimize the effect of session order on the results. Participants filled in a questionnaire at the end of each session. They were given a 2-minute break between the two sessions to relieve any discomfort that may have arisen from the previous session. We concluded the experiment with a debriefing and asked participants if they had any comments to add. The experiment lasted approximately 55 min in total. The experiment procedure is illustrated in Figure 6b.

6.3. Participants

Twenty participants (11 females and 9 males) took part in the visit experiment, aged between 19 and 26 ($M = 21.25, SD = 1.92$). We have thirteen participants from Study 2 rejoining the study.

Table 1. Measurement items of the constructs, the factor loading, and measurement validity and reliability results, showing the cronbach's alpha (CA), composite reliability (CR), average variance extracted (AVE), and the square root of AVE of each construct. PVQ = Photograph and video quality, AM = autobiographical memory, PE = positive emotion, RI = revisit intention.

Construct	Item	Description	Loading	CA	CR	AVE	PVQ	AM	PE	RI
PVQ	PVQ1	The photographs and videos in personal memory museums were unique.	0.680	0.776	0.908	0.685	0.828			
	PVQ2	The photographs and videos in personal memory museums were very interesting.	0.935							
	PVQ3	The photographs and videos in personal memory museums were very affluent.	0.847							
AM	AM1	The photographs and videos in personal memory museums could remind me of previous memories.	0.686	0.806	0.802	0.633	0.299	0.795		
	AM2	The photographs and videos in personal memory museums proved past memories.	0.853							
	AM3	The photographs and videos in personal memory museums were evidence of past memories.	0.844							
	AM4	The photographs and videos in personal memory museums were important because they carried the memory of the past.	0.788							
PE	PE1	I am very relaxed during my visit to personal memory museums.	0.665	0.731	0.762	0.656	0.309	0.570	0.810	
	PE2	I am very proud of the visit to personal memory museums.	0.881							
	PE3	I am very excited about the visit into personal memory museums.	0.865							
RI	RI1	I am likely to visit the personal memory museums again.	0.925	0.892	0.893	0.822	0.527	0.610	0.685	0.907
	RI2	If there is a chance, I would stay longer in personal memory museums.	0.901							
	RI3	I am likely to visit the personal memory museums again in the future.	0.894							

Participants were undergraduate ($n = 16$) and postgraduate students ($n = 4$). Unlike participants in Study 1 (refer to Sec 3.2), who were slightly familiar with VR, the majority of participants in this study had prior experience with VR (18/20, 90%). They were very familiar with computer ($M = 4.00$, $SD = 0.97$) and familiar with 3D graphics ($M = 3.25, SD = 1.02$) and VR ($M = 3.45$, $SD = 1.28$). Participants reported that they enjoyed collecting and sharing personal memories in their daily lives ($M = 3.35, SD = 0.81$).

6.4. Analysis and results

We used IBM SPSS Statistics to conduct descriptive analysis and also comparative analysis to identify the differences between PC and VR visits. Shapiro-Wilk tests were conducted to examine the data distributions. For data that conformed to a normal distribution, we adopted a paired samples t-test. For data that did not fit a normal distribution, we conducted the Wilcoxon signed-rank test. In addition, we used SmartPLS 4 to validate the research model.

6.4.1. Usability and user experience

The mean SUS scores of the PC and VR visits were 76.75 ($SD = 8.43$) and 79.88 ($SD = 11.99$), respectively. Both values exceeded the threshold value of 68.00, indicating satisfactory usability of the MemoryVR system in both visits. A paired samples t-test showed no statistically significant difference in usability between PC and VR, $t(19) = -1.296, p = 0.211$.

The UEQ results are shown in Figure 7b. The values of pragmatic quality, hedonic quality, and overall experience in PC visit and VR visit exceeded the suggested value of 0.8 (Laugwitz et al., 2008), which were 1.85 ($SD = 0.78$), 1.68 ($SD = 0.84$), and 1.76 ($SD = 0.72$) for PC visit, and 1.75 ($SD = 0.84$), 1.95 ($SD = 0.88$), and 1.85 ($SD = 0.72$) for VR visit, respectively, indicating that users had a positive experience visiting others' personal VMs. Comparing our results with the UEQ benchmark dataset (Laugwitz et al., 2008), two visit modes were of *Excellent* (top 10%) user experience in terms of pragmatic quality, hedonic quality, and overall experience.

Paired sample t-tests showed no statistically significant difference between PC and VR visits in either the pragmatic quality ($t(19) = 0.639, p = 0.530$) or the overall experience ($t(19) = -0.687, p = 0.500$). No statistically significant difference was found in hedonic quality between them either, $Z = -1.657, p = 0.098$.

6.4.2. Presence

A Wilcoxon signed-rank test showed significant differences in perceived presence between PC and VR visits, $Z = -3.503$, $p < 0.001$. Participants had a greater sense of presence using VR ($M = 6.25, SD = 0.72$) than using PC ($M = 4.60, SD = 1.27$).

6.4.3. Time spent and selection of VMs

6.4.3.1. Time spent. The time duration of PC visit varied from 112s to 1048s ($M = 501.85$, $SD = 273.46$), and the time range of VR visit was 224s to 1324s ($M = 620.95, SD = 349.89$). There was no statistically significant difference in time spent between PC and VR ($z = -1.587, p = 0.113$).

6.4.3.2. Selection of VMs. Participants visited 2 to 11 VMs in the PC visit ($M = 4.90, SD = 2.36$) and selected 2 to 9 VMs in the VR visit ($M = 4.75, SD = 2.00$). Most of them selected different virtual museums to visit using PC and VR. The reasons they selected the VMs can be categorized into six dimensions: content described ($n = 61$), following the list order ($n = 54$), recognizing a friend's VM ($n = 44$), being curious about the naming ($n = 22$), revisiting the same VM ($n = 8$), and visiting a VM of their own ($n = 4$). Participants mostly chose from the content described. For example, when VP1 chose to visit the VM named "Did you see the concert today?," she said, "I've been craving a concert lately, so let's see what's in this concert VM first." Taking sequential visits according to the list order appeared to be another reason. "I don't know what to pick, so I'll see them one by one." (VP12). Because of our snowball sampling approach, many participants invited their friends to join the study. Thus, visiting friends' VMs became an important reason for their choices. In addition, participants also visited VMs with names that they were curious about. There were also participants who chose to revisit the same VM because they liked it, or they wanted to compare the experience using a PC and VR.

6.4.3.3. Snapshots. Seven participants took snapshots during the PC visit. The number of snapshots they took ranged from 1 to 5 ($M = 2.71, SD = 6.6$). During the VR visit, ten participants took photos, and the number ranged from 4 to 22 ($M = 8.60, SD = 6.06$). Figure 9 includes some examples.

6.4.4. User revisit intention

Figure 10 presents the box plots of the URI scores. Wilcoxon signed-rank tests showed no statistically significant differences in *Photograph and Video Quality* ($Z = -0.697, p = 0.486$), *Autobiographical Memory* ($Z = -1.749, p = 0.08$) or *Positive Emotion* ($Z = -0.550, p = 0.582$) between PC and VR. However, there was a statistically significant difference in *Revisit Intention* between them ($Z = -2.203, p = 0.028$). Users had a greater revisit intention using VR than using a PC.

6.4.4.1. Reliability and validity analysis. Table 1 shows the results of the reliability and validity analysis of our research model. Specifically, the values of Cronbach's alpha (CA) ranged from 0.731 to 0.892, exceeding the threshold value of 0.7 (Fornell & Larcker 1981), thus demonstrating a high measurement reliability. The values of composite reliability (CR) ranged from 0.762 to 0.908, above the standard value of 0.7, which indicated sufficient internal consistency. In addition, the average variance extracted (AVE) ranged from 0.633 to 0.822, which was greater than the threshold value of 0.5 (Fornell & Larcker 1981), indicating that convergence validity was confirmed. For discriminant validity (Fornell & Larcker 1981), the square root of AVE of each construct was greater than its correlation with other constructs (see the last four columns of Table 1), demonstrating acceptable discriminant validity.

6.4.4.2. Structural equation modeling analysis. Figure 10 includes the detailed statistics of the path coefficients and the hypothesis testing results. Our analysis showed an R^2 value of 0.541 (> 0.5)) for *Revisit Intention*, implying the structural model has moderate predictive power (Hair et al., 2011). In addition, the Q^2 value of *Revisit Intention* was 0.411 (> 0), indicating the research model has acceptable predictive relevance (Leguina, 2015). With the confirmation of the proposed hypotheses, the results showed that *Photograph and Video Quality* has significant positive effects on *Autobiographical Memory* (H1); *Autobiographical Memory* has a significant positive impact on *Positive Emotion* (H2); and *Autobiographical Memory* and *Positive Emotion* have significant positive impacts on *Revisit Intention* (H3, H4).

6.5. Summary and discussion

6.5.1. Participants' behavior and expectation when visiting MemoryVR

Visits in both PC and VR were well received by participants, indicated by their usability and user experience ratings. From the user comments and our observations, we found that the sense of ownership and curiosity have contributed to their engagement in MemoryVR. Participants mentioned how they liked to visit VMs created by themselves. For example, *I feel excited and fulfilled when I see that the museum I arranged with my own photos can be successfully browsed in VR* (VP18). This underscores

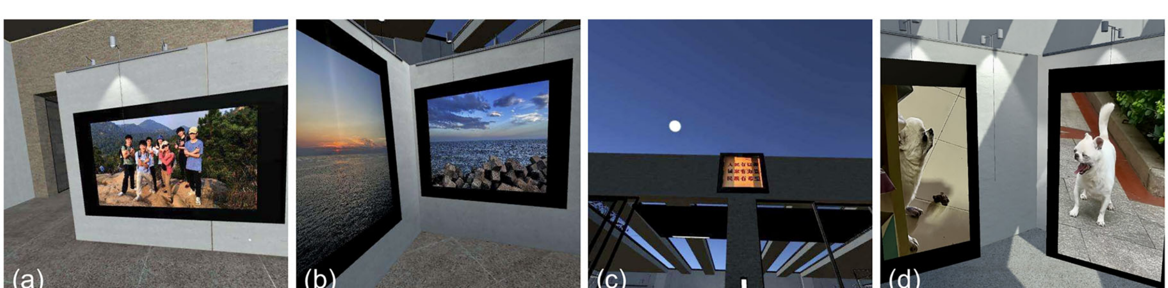
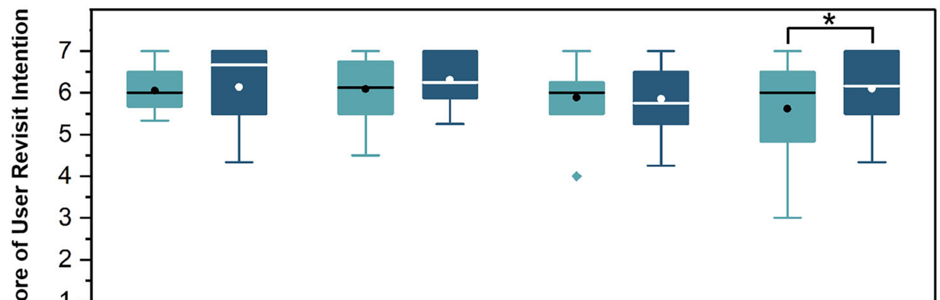


Figure 9. Some snapshots and photos taken by participants, showing content in the themes of (a) close relationship, (b) travel, (c) decoration, and (d) pets.



	PVQ	AM	PE	RI
PC	6.05 (0.58)	6.09 (0.80)	5.89 (0.81)	5.62 (1.11)
VR	6.13 (0.91)	6.31 (0.62)	5.85 (0.78)	6.10 (0.82)

Figure 10. Model validation results. Values on the path are coefficients (β). $*p < 0.05$; $**p < 0.01$; $***p < 0.001$. Box plots and the means (with standard deviations) show the results of the user revisit intention measures. PVQ = Photograph and video quality, AM = autobiographical memory, PE = positive emotion, RI = revisit intention.

the substantial value of engaging users in content creation and supporting the presentation of user-generated content. MemoryVR goes beyond passive viewing of narratives crafted by others, empowering users to become active creators who share their own stories and cultivate a sense of ownership.

Although participants in Study 1 reported that they were willing to share their personal VMs with a specific group, especially their close friends and family, we observed an interesting phenomenon in the laboratory experiments: participants showed strong curiosity when visiting others' VMs, including those created by individuals they were not acquainted with. Even when certain areas appeared empty, participants showed persistent curiosity, with comments like, "What if there is something hidden by the owner behind this display board?" (as mentioned by VP12). They actively searched for hidden surprises or unexpected elements, especially within the VR scenes. Curiosity also extended to the naming of the VMs. When confronted with unfamiliar VM names, rather than losing interest, participants became more intrigued. This inherent curiosity and spontaneous exploration appeared to be common among participants and likely contributed to their expectations for re-experiencing the past.

Regarding access to personal VMs, most participants in Study 2 were willing to make their personal VMs publicly available. However, this is likely because they selected memory moments that were appropriate to share with others when they agreed to participate in the experiment in the first place. A number of participants expressed that if it were not an experimental setting, they might upload different content, including more private photos, and restrict access to only themselves or the very close ones. This highlights the consideration of privacy and content selection in sharing personal memories within digital platforms.

Despite the overall positive feedback, we also observed that many participants have attempted to pick up the picture frames or sculpture decorations during their VR visit. Upon realizing that they could not interact with them, some participants showed a slight sense of disappointment. Enhancing support for perceived affordances is an area for future improvement.

6.5.2. Revisit intention of personal virtual museums

It is widely accepted that revisit intention is useful to predict actual revisit behavior (Jang & Feng, 2007). We validated a research model based on a previous study on visitors' revisit intention in tourism (Zhang et al., 2021) and showed that it is valid for predicting users' revisit intention of MemoryVR, which was a personal VM system. In that study, Zhang et al. (2021) concluded that travel photos can be used as a medium to transform tourists' continuous shaping of self-concept into autobiographical memories. Previous studies also showed the positive effect of sensory experiences on the reproducibility of user memory (Agapito et al., 2017). In our study, we found that users acknowledged the clarity and immersion in VR, and found it to have contributed to their perceived presence compared to the PC display. Participants also suggested that increasing the image resolution and enriching the supported file types could be a potential improvement in the quality of photos and videos. In addition to aspects of media characteristics, a higher degree of customizability in content could also improve the perceived sense of uniqueness and affluence within the personal VMs. These could promote users' autobiographical memories.

Additionally, the positive effect of autobiographical memory on positive emotion suggests that feeling related to others' memories and being reminded of one's past experiences is likely to evoke positive emotional responses. Participants mentioned that they enjoyed the nostalgic feeling generated by recalling past achievements, life moments, and time spent with friends and families during the MemoryVR experience. In addition, they also made some interesting comments when they visited others' VMs. Introducing features that allow participants to react to others' memory moments, such as the like and comment features on social media, could potentially create new memory experiences and further enhance positive emotions. Autobiographical memory and positive emotion were also shown to be significant antecedents of users' revisit intention. Participants were engaged in their visits and showed great curiosity in exploration. They reported an *Excellent* user experience for both PC and VR visits, and no significant differences were found between them. However, they showed a significantly greater revisit intention using VR. The characteristics of our sample indicated that the majority of participants had previous experience with VR and were familiar with the technology, which suggests that novelty is unlikely to be a major factor. This is more likely because VR supports the immersive experience of memories and allows participants to revisit the past. The perceived presence may have contributed to their feeling of connection with others' memories, and the immersive autobiographical experience positively affected their revisit intention. Furthermore, the results demonstrated that positive emotion was an important motivation factor for revisit intention, which is consistent with the findings of Del Bosque and San Martín (2008). Therefore, designers should assist users in generating positive emotions such as relaxation, satisfaction, pride, and excitement in order to increase their revisit intentions.

Users in our study expressed feelings of ownership, curiosity, and meaningful connection while visiting both their own and others' virtual museums. Despite that the current study excluded place attachment constructs to have a clear focus on autobiographical memory, we acknowledge that immersive virtual environments can foster a sense of place, particularly through the sense of spatial presence, customization, and emotional resonance from users' behaviors. Future work could explore the effect of virtual place identity and dependence on user emotion and revisit intention. These could have a potential influence on determining their role in long-term revisit behaviors and emotional resonance.

7. Design recommendation

To understand the user experience and their behavior in MemoryVR, we have conducted two exploration user studies to answer three RQs in Section 5 and Section 6. Based on the findings and discussions, we generated five types of design recommendations (DRs) for the relevant content in personal VMs in order to provide a better user experience.

7.1. DR1: Include common devices such as PCs if possible to accommodate a wider range of users when setting a personal VM

The results from Study 1 showed that though the acceptance of VR is generally good, the number of people who own their own VR devices today is still in the minority. This demonstrates that the range of users tends to decrease, and the cost tends to increase if VM only supports the use of VR devices. Hence, it is beneficial to add the suggested commonly used devices to design the VM. The familiarity of people with commonly used devices can be effective in mitigating the learning barriers associated with new technologies (Bringman-Rodenbarger & Hortsch, 2020). Integration with PCs could facilitate access to the VM for users who may not have the latest devices and promote inclusivity by providing access.

7.2. DR2: Set accessibility permission to clarify privacy perception and protection

The results gleaned from questions 25-26 in Study 1 indicated that the majority of participants were willing to share their memories with specific groups (e.g., close friends and family). The participants in Study 2 also noted that they would choose different uploaded content if their personal VM were demonstrated to different audiences. Accessibility, as one of the key concepts of privacy on social media, has been widely discussed (Chen & Cheung, 2018). As a mechanism for determining who can access and to what extent this information is accessible, it is important for individuals to maintain relationships with others. Hence, it is recommended to implement the setting of access permission for personal VMs.

7.3. DR3: Incorporate text, audio, and 3D models into upload types to enrich users' choices

We found that multiple users mentioned the need to include text, audio, and 3D models in their practical process, though these media did not show a high profile in the Study 1 survey. As the important message types in their daily posts on the social media platform (Shahbaznezhad et al., 2021), the text and audio should be considered in the personal VM setting. In recent years, users can effectively create photo-realistic 3D models of objects based on high-quality 3D scanners and close-range photogrammetry techniques (Ulvi, 2021). From museums replicating parts of cultural heritage (Li & Ch'ng, 2022), to an international student scanning and modeling her own dormitory scene for commemoration (Anonymous, 2024), 3D models have demonstrated their unique advantages in visualization and nostalgia. Thus, we recommend incorporating 3D models into the personal VM settings.

7.4. DR4: Provide semi-customized templates from the perspective of different themes to cater to personal characteristics

Since all creations in Study 2 were based on the same initial scene, some participants were frustrated by the inability to change the default layout of the gallery. Similarly, during the visit process (in Study 3), participants expressed that different scenes were too similar, resulting in a lack of personal characteristics and styles. Therefore, according to the contents uploaded to their personal VM (see Section 5.4.2), we recommend providing users with a variety of styles of templates, with themes that can prioritize close relationships, travel, decoration, pets, performance, food, achievement, and the outdoor.

7.5. DR5: Give more interactive settings during the visit

The results of Study 3 showed the participants' expectations for a more interactive design during the visits. Interaction is pivotal in creating an engaging virtual environment as it offers the essential affordances that transform static virtual objects into interactive ones (Li & Ch'ng, 2022). Basic tasks such as selecting, rotating, and scaling 3D models in the VM context have been researched (LaViola et al., 2017; Wang et al., 2023). It is necessary and feasible to introduce these basic interactions into personal VMs, and more complex interactions that align with people's mental models can also be explored. Additionally, designers should offer the option of reset to help owners restore the original scene (Shneiderman et al., 2016).

8. Limitations and future work

This research has some limitations and presents avenues for future work. First, the limitations in sample characteristics should be noted. Our Study 1 sample size was reasonably large, but all participants appeared to be Chinese, and most of them were from relatively developed areas. It is important to address regional limitations and refine demographics in future research to ensure broader representation. In addition, we had a participant pool of young adults in Study 2 and Study 3. Given that the hippocampus, a key element in learning and memory, is susceptible to aging (Bartsch & Wulff, 2015; Glatigny et al., 2019), it is essential to investigate whether older individuals exhibit similar attitudes toward MemoryVR as younger ones.

Second, researchers generally agree that memory plays a central role in the experience of duration in retrospect (Glatigny et al., 2019). However, our studies primarily evaluated participants' immediate experiences, leaving the long-term effects of MemoryVR on users unexplored. Future research should consider conducting longitudinal studies to examine the impact of MemoryVR on memory experiences over time.

Third, to avoid the potential bias stemming from participants' familiarity with technology, particularly regarding 3D selection and manipulation, our study was conducted using a standard PC rather than immersive media in a 3D gallery format. In this setup, users primarily placed and viewed 2D images and videos within the virtual space, as these are the most commonly used mediums for preserving memories. This design decision simplified the experience, reducing complexity and allowing us to focus on exploring users' behaviors and expectations in this context. Nonetheless, future versions of MemoryVR could incorporate more immersive features, such as 360° video (Xu et al., 2020), ambient audio (Zimmermann & Lorenz, 2008), and contextual 3D reconstructions (Yao & Li, 2024). These novel media can provide an immersive experience that is proliferating into our daily lives and is gradually becoming an important type of VR content (Cai et al., 2018; Xu et al., 2020). Investigating how immersive media can enhance the user experience of memories represents a valuable direction for future research.

Fourth, while MemoryVR asserts the benefits of storing and sharing memory, it is important to recognize that forgetting is a crucial aspect of memory. Researchers have shown that forgetting is an adaptive mechanism for restricting the influence of past experiences on present experiences (Gemmell et al., 2006). This mechanism can limit memories that are emotionally and self-relevant but inconsistent with the present self (Stajano & Anderson, 2002). In the field of digital memories, forgetting is also relevant for reducing privacy risks in ubiquitous computing as well as improving the practicality of storing information (Davies et al., 2015). Nevertheless, there is limited understanding of how these digital memories ought to be embedded and managed. We expect that there will be more ethical and technical support for addressing these aspects of memory studies in the future.

Finally, another limitation pertains to our measurement of autobiographical memory. We adopted a scale validated by Zhang et al. (2021) to maintain methodological consistency. However, this measurement is limited to capturing the distinctive characteristics of autobiographical memory, particularly those emphasizing narrative identity and emotional nuances (Kirkegaard Thomsen et al., 2025). Future studies should consider developing and validating measurement items that better reflect these psychological dimensions of autobiographical memory.

9. Conclusion

In this paper, we present three studies around digital memories and personal virtual museums. Our first survey study contributes to the understanding of the way users record, store, and share digital memories, and their expectations of the use of immersive virtual reality in their memory experience (**RQ1**). Based on the results, we specified some key functions and constraints that informed the design of MemoryVR, a personal virtual museum system that supports the use of PC and VR. It allows individuals to organize and share their memories in a virtual museum environment and to visit the memory spaces of others in an immersive way. We conducted two laboratory studies to evaluate MemoryVR and investigate users' experience and behavior when they create (**RQ2**) and visit **(RQ3a)** by MemoryVR. Overall, participants provided high ratings for the pragmatic and hedonic qualities of the user experience when visiting personal VMs. Observations and qualitative analysis showed that MemoryVR was perceived as a ritualistic and fulfilling experience, evoking a sense of ownership, curiosity, and user engagement. In addition, we explored the factors influencing user revisit intention of personal virtual museums (**RQ3b**). The results showed that the photograph and video quality have a positive effect on users' autobiographical memory, which in turn influences their positive emotions. These two factors positively affect users' revisit intention. A discussion of limitations and future work provides researchers with insights into the design of customized 3D VMs for enhanced user experience of digital memories.

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Appendix A. survey questions used in study 1

Questions	Results
Q1. Are you willing to participate in the research?	Yes = 204, No = 4
Q2. Gender	Female = 106, Male = 96, Other = 2
Q3. Age	M = 27.81. SD = 9.36
Q4. Job	Student = 95, Office Staff = 65, Professional = 2, Business people = 14, Others = 28
Q5. Education background	Middle School = 1, High school = 16, College's degree = 32, Bachelor's degree = 126, Master's degree = 23, Doctor's degree = 6
Q6. How do you like to record your life?*	Smartphones = 180, PC and Laptops = 70, Camera, Polaroid, etc.=71, Non-digitally on a notebook = 24, Others = 4
Q7. What are the digital memory formats would you like to keep?*	Image = 167, Video = 159, Text = 95, Voice = 59, Others = 4
Q8. What would you **record** in your life?*	Achievement = 113, Work = 101, Performance = 53, Food = 135, Travel = 157, Outdoor = 113, Pets = 115, Close friends, Family, Colleagues, etc.=121, Others = 1
Q9. How often would you **record** your memory moments?	M = 3.28, SD = 1.07 (from 1 = Never to 5 = Everyday)
Q10. How do you like to **view** your photos and videos in the past?*	View them on a digital device = 192, Print them out in a photo album = 53, Others = 1
Q11. How would you **organize** your photos and videos in the past?*	Photo albums in mobile phone = 150, Folders in computer = 117, Cloud service (e.g., Baidu Cloud, DropBox, etc.)=93, Others = 5
Q12. How often would you **check** your memory moments in the past?	M = 2.57, SD = 0.84 (from 1 = Never to 5 = Everyday)
Q13. How satisfied are you with your current **storage** of personal memory?	M = 3.36, SD = 0.83 (from 1 = Not at all to 5 = Extremely)
(If not satisfied) Q14. Can you briefly explain what you are not satisfied with?	See examples in article.
Q15. How satisfied are you with your current **viewing** of personal memory?	M = 3.44, SD = 0.81 (from 1 = Not at all to 5 = Extremely)
(If not satisfied) Q16. Can you briefly explain what you are not satisfied with?	See examples in article.
Q17. How satisfied are you with your current **sharing** of personal memory?	M = 3.52, SD = 0.72 (from 1 = Not at all to 5 = Extremely)
(If not satisfied) Q18. Can you briefly explain what you are not satisfied with?	See examples in article.
Q19. How would you like a 3D virtual space that hosts a collection of photos and videos?	See results in article.
Q20. If you were asked to create a personal museum (a 3D virtual space) that hosts your memory, what kind of media would you like to include?*	See results in article.
Q21. How often do you **share** your selected life moments (e.g., texts, photos, videos) online (e.g., WeChat, TikTok, Little Red Book, Linkedin, Facebook, etc.) publicly?	M = 2.49, SD = 0.94 (from 1 = Never to 5 = Everyday)
Q22. How often do you **share** your selected life moments (e.g., texts, photos, videos) online (e.g., WeChat, TikTok, Little Red Book, Linkedin, Facebook, etc.) with close friends, family, colleagues?	M = 3.00, SD = 1.11 (from 1 = Never to 5 = Everyday)
Q23. How often do you **view others'** selected life moments online (e.g., TikTok, Little Red Book, Bilibili, Linkedin, Facebook, etc.)?	M = 3.47, SD = 1.23 (from 1 = Never to 5 = Everyday)
Q24. How often do you **view your close friends, family and colleagues'** life moments online (e.g., WeChat moments, TikTok, Little Red Book, Bilibili, Linkedin, Facebook, etc.)?	M = 3.52, SD = 1.24 (from 1 = Never to 5 = Everyday)
Q25. If you were asked to create a personal museum (a 3D virtual space) that hosts your memory, who would you like to share this personal museum with?	See results in article.
(If specific group) Q26. Please choose all the specific groups that you may share your personal museum with.*	See results in article.
Q27. Have you ever used Virtual Realiuty (VR) applications?	Yes = 132, No = 72
Q28. Please evaluate your knowledge of VR.	M = 2.53 SD = 1.14 (from 1 = Not at all to 5 = Extremely)
Q29. Do you have any VR equipment?	Yes = 54, No = 150
(If yes) Q30. Which VR equipment do you have?*	Oculus Quest = 29, HTC Vive = 30, Pico = 16, Others = 0
Q31. Please evaluate your interest in experiencing new technologies.	M = 3.22, SD = 1.16 (from 1 = Not at all to 5 = Extremely)
Q32. How willing are you to invest in learning new things?	M = 3.95, SD = 0.84 (from 1 = Not at all to 5 = Extremely)
Q33. How willing are you to build your personal museum (a 3D virtual space) that hosts your memory by your own?	M = 3.63, SD = 1.00 (from 1 = Not at all to 5 = Extremely)
Q34. What is most likely to drive you to build a personal museum (a 3D virtual space) that hosts your memory?	Re-experience the past = 175, Storage = 141, Social function = 113, Others = 5
Q35. Are you willing to participate in the follow-up lucky draw?	Yes = 85, No = 119
Q36. If we later develop a virtual 3D museum that can be edited by yourself to re-experience personal or other people's memories, would you like to know more or participate in system testing?	Yes = 81, No = 123

*Multiple-Choice Questions.